Job Market Analysis

Project: **PRDA – 04 Job Market Analysis**

Tools Used: MySQL, Power BI, Excel

Dataset: project_job_market_analysis

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Project Overview

The Job Market Analysis project focuses on analysing job postings related to Data Science and other related roles. The project aims to understand job availability, salary trends, industries, companies, job titles, education requirements and technical skills.

SQL was used for data extraction, cleaning, transformation and analysis, while Power BI was used to create interactive dashboards and visualize the results.

Objective

The Data Science job market contains a large number of job postings with differences in salaries, locations, industries, companies, job titles, education requirements and technical skills. The purpose of this project is to analyse these factors and identify meaningful patterns from the available job-posting data.

1. Analyse states with the highest number of job postings.
2. Calculate average minimum and maximum salaries by state.
3. Analyse average salary across states.
4. Identify the top industries.
5. Identify companies with maximum job openings.
6. Identify the most common job titles.
7. Analyse salary by job title.
8. Analyse the relationship between salary and education.
9. Analyse additional relationships between available features.

Dataset Description

The dataset consists of job-market records containing information about job titles, salary estimates, companies, locations, ratings, industries/sectors, education-related information and technical skills.

The Power BI dashboard contains 12K job postings, 343 companies, and 60 industries. The overall average salary displayed in the dashboard is 101.48. The dataset contains customer service call records and includes information such as:

- Customer ID
- Customer name
- Sentiment
- CSAT score
- Call timestamp
- Call day
- Reason for calling
- City , State
- Communication channel
- Response time
- Call duration
- Call centre

Tools Used

MySQL was used for:

- Data validation
- Record analysis
- NULL-value checking
- Duplicate checking
- Category analysis
- Aggregation and statistical analysis

Power BI was used for:

- Data transformation
- Dashboard creation
- KPI cards
- Charts
- Maps
- Interactive analysis

Data Preparation

The data was initially retrieved using MySQL. The Market table was used for analysis. Data was examined for missing values, data types and unique values. Required fields were transformed into appropriate formats for analysis.

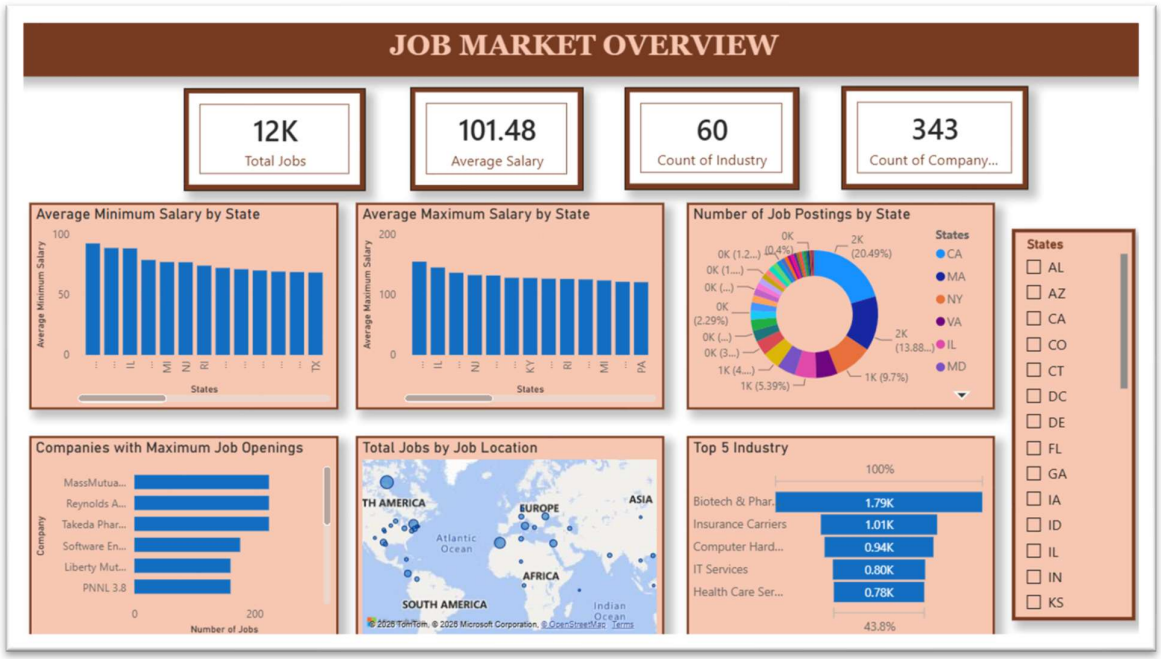
Power BI was used for Data Transformation and to create two dashboard pages.

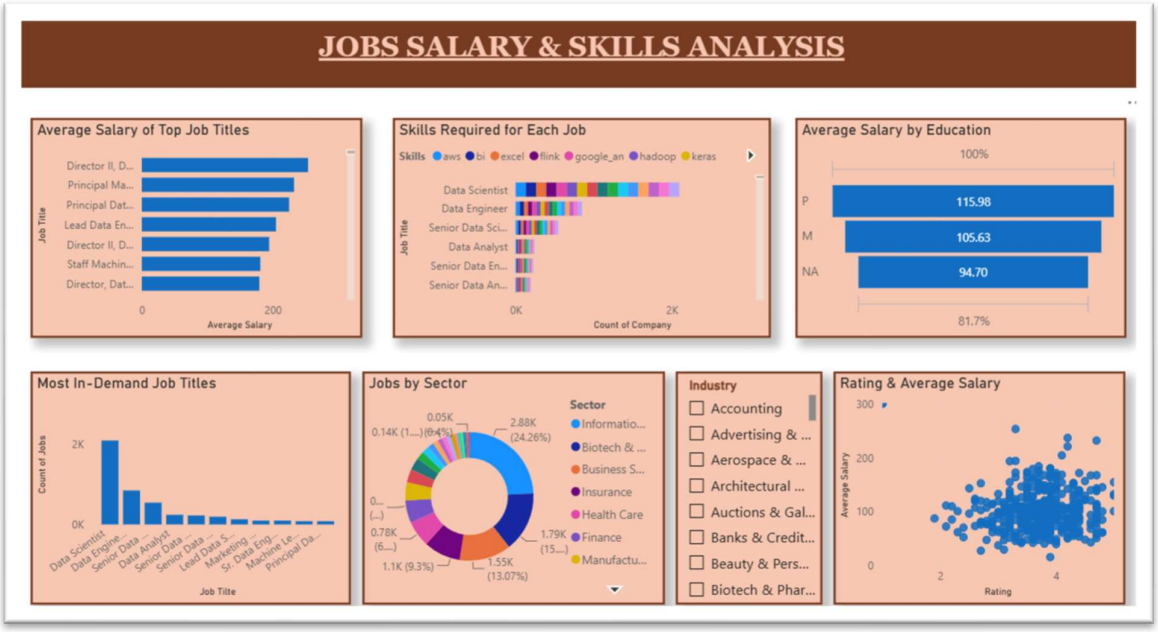
Data Tranformation:-

- Verified column data types
- Converted the call timestamp field to Date
- Checked data quality
- Checked for errors
- Verified that there were no NULL values in the analysed fields
- Prepared the data for visualization

Dashboard:-

The first page focuses on the overall job market, while the second page focuses on salary and skills analysis.





Key Insights

- The dataset contains approximately 12K job postings.
- The average salary displayed in the dashboard is 101.48.
- The dataset contains 60 industries and approximately 343 companies.
- Data Scientist is the most frequently occurring job title in the displayed job-title analysis.
- Biotech & Pharmaceuticals has the highest number of postings among the top five industries displayed.
- Job postings and salary levels vary across different states.
- Average salary differs across job titles and education categories.
- Different job titles require different combinations of technical skills.
- The rating and salary visualization shows salary distribution across different company ratings.

Suggestions / Recommendations

- Companies can analyse the distribution of job postings and salaries to understand hiring patterns across different locations.
- Organizations can identify commonly required technical skills when designing job roles and training programs.
- Job seekers can use information about frequently occurring job titles and technical skills to understand the skills commonly appearing in the analysed job postings.
- Organizations can compare salary levels across job titles and locations when analysing compensation patterns.

Conclusion

This project analysed the Data Science job market using SQL and Power BI. The analysis covered job postings, locations, salaries, companies, industries, job titles, education and technical skills.

The interactive dashboard provides a consolidated view of the analysed data and makes it easier to identify differences in job availability, salary and skill requirements across various categories.

The project demonstrates the use of SQL for data analysis and Power BI for interactive data visualization and reporting.